

◆ O-Level learning, one web at a time

THE WEB BEHIND THE WEB.

The complete technical picture of **EduNets** — the spaced-repetition study platform that tracks what students are forgetting before it costs them the exam.

App **EduNets (StudyNow)**

Build **appbuilder 0.2.0**

Stack **React 19 · Vite 7 · TS 5.9**

Doc date **11 Aug 2026**



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12 minutes today beats 12 hours the night before finals.

EduNets is a single-page web app for Singapore O-Level students. It models each topic as a node in a personal **concept web**, scores how well the student remembers it, decays that score over time using the Ebbinghaus forgetting curve, and nudges the student — and their friends — to rescue topics before they slip. A parallel **teacher view** turns the same data into class-level insight. Spidey the spider mascot (with a bee companion) guides both.

Smart Quiz

Syllabus-mapped questions that show what to review next. Every attempt updates the topic's Memory Score and schedules the next review.

Concept Web

An interactive, pan-and-zoom bubble graph of every topic. See what connects — and where the gaps are — with a one-tap weak-topics filter.

Ask Teacher

Students turn a stuck moment into a direct question; teachers triage replies from a dedicated inbox and dashboard.

Two personas, one codebase

A global `viewModeAtom` (Jotai) switches the entire app between **Student** and **Teacher** modes. Students get quizzes, squads and rescues; teachers get Class Pulse, ranked actions (deliberately capped at five urgent items), a class-wide concept web and quiz review.

A client-first SPA with a Power Platform bridge.

EduNets ships as a fully client-side React application. Demo data and user progress live in browser storage via persistent Jotai atoms; the Microsoft Power Apps SDK connects the same app to Dataverse when deployed inside Power Platform.

Frontend

- **React 19 + TypeScript 5.9**, built with **Vite 7**
- **Tailwind CSS v4** with a custom EduNets theme layer (`src/index.css`)
- **Radix UI / shadcn** component library — 40+ primitives in `src/components/ui`
- **motion** for animation, **lucide-react** icons, **recharts** for charts, **sonner** toasts

State & data

- **Jotai** atoms for app state; `atomWithStorage` persists progress locally
- **TanStack Query** for async data fetching and cache
- **React Router 7** client-side routing with a shared sidebar layout
- **@microsoft/power-apps** SDK — `initialize()` on boot; Dataverse tables (e.g. `cree0_student`) configured in `power.config.json`

Build & quality pipeline

Typecheck

`tsc` / `tsgo` strict, no emit



Lint

`oxlint` quiet mode



Bundle

Vite 7 production build to `dist/`



Deploy

Vercel (web) or Power Platform (`power.config.json`)

Package managers: bun + pnpm workspaces. `check:fast` runs typecheck, esbuild smoke-bundle, lint and Tailwind compile concurrently.

Fourteen routes, one shared layout.

`/welcome` and `/signup` render standalone; everything else nests inside `_layout.tsx`, which mounts the app sidebar, mascots and view-mode switcher.

ROUTE	PAGE	PERSONA	WHAT IT DOES
<code>/welcome</code>	Welcome	Public	Marketing hero + 5-question onboarding quiz (role, school, starting material, subjects, confidence). Answers stored in <code>sessionStorage</code> , then routed by role.
<code>/signup</code>	Sign up	Public	Account creation flow.
<code>/</code>	Dashboard	Student	"EduNets study pulse" — Memory Score overview, at-risk topics, teacher-reply alerts, quick actions.
<code>/quiz</code>	Smart Quiz	Student	Configure subject/topic, auto-set difficulty, answer syllabus-mapped questions, compare attempts, get a spaced-repetition schedule.
<code>/concept-web</code>	Concept Web	Both	Router splits to the student web or the teacher class web based on view mode.
<code>/study-squad</code>	Study Squad	Student	Squad roster, group streak, referral code, Memory Score recap, "EduNets Wrapped" weekly stats.
<code>/rescue-join</code>	Rescue join	Student	Entry point for invited friends to join an active rescue.
<code>/rescue-room</code>	Rescue Room	Student	Real-time multiplayer quiz rounds with locked answers, live ranks and streak credit.
<code>/capture-hub</code>	Capture Hub	Student	Upload notes (PDF, Word, PowerPoint, OneNote, images) with OCR extraction, or import a Teams meeting recording.
<code>/ask-teacher</code>	Ask Teacher	Student	Send a question straight to a chosen teacher, with optional subject tagging.
<code>/teacher-inbox</code>	Teacher Inbox	Teacher	Triage and reply to student questions.
<code>/teacher-dashboard</code>	Teacher Dashboard	Teacher	Urgent items (capped at five), ranked actions sorted by urgency, Class Pulse per class.
<code>/profile</code>	Profile	Both	User settings and achievement badges.
<code>*</code>	Not found	Public	Friendly 404 in brand style.

How the web stays alive.

Memory Score engine

Every topic carries a base score set by quiz performance. An **Ebbinghaus forgetting-curve decay** (`calculateDecay` in `study-data.ts`) subtracts points for each day since the last review, so the *effective* score a student sees is `max(0, base - decay)` . Anything below **42** is flagged "at risk" and surfaces across the dashboard, concept web and squad nudges. Each quiz result maps a percentage to a score tier and schedules the next review at 9:00 AM on a spaced-repetition date.

< 40%
score 28

≥ 40%
score 45

≥ 55%
score 62

≥ 75%
score 80

≥ 90%
score 95

Question bank

A 1,700-line typed bank (`question-bank.ts`) mapped to real Singapore O-Level syllabuses, with per-question mark schemes and keyword hints that explain wrong answers.

Biology 5090

Chemistry 5070

Physics 5054

Combined Sci 5129

History 2176

Geography 2236

English 1128

A-Math 4049

E-Math 4048

- Formats: `mcq` , `structured` , `essay` across Paper 1 / Paper 2
- Difficulty: foundation → intermediate → advanced, auto-picked from the current Memory Score

Study Squad & Rescue Room

Squads make forgetting social. Members see each other's overall scores, keep a **group streak**, and send **rescue nudges** when a friend's topic drops.

- Rescues open a multiplayer quiz room (`QuizRoomDraft`) with rounds, live scores, avatar colours and final ranks
- Completed rescues log a `RescueActivityCompletion` and can restore the streak
- Referral codes (e.g. `EDUNETS-5DAY`) and an "EduNets Wrapped" weekly recap drive retention

Concept Web

The signature view: topics as bubbles sized and coloured by effective score, linked by prerequisite edges. Click any bubble; drag to pan, scroll to zoom.

- Student web shows personal scores plus **friend markers** on shared weak nodes
- Teacher web aggregates the class and highlights common gaps
- "Show weak topics only" filter isolates the at-risk subgraph

Capture Hub & teacher tools

Material in, insight out. Students upload notes with OCR text extraction or import Teams recordings; captured material feeds topic creation.

- Teacher Dashboard: urgent items capped at five "so it stays actionable", ranked cross-feature actions, Class Pulse
- Teacher Inbox + quiz review close the loop on Ask Teacher
- Gamification: 8 achievements (Quiz Champion, Streak Master, Mind Mapper, Note Collector, Perfect Score, Topic Master, Renaissance Student, Early Bird) plus animated Spidey and bee companions

Typed to the last topic.

All domain types live in `src/Lib`. The shapes below are the backbone; demo fixtures (six squad members, per-topic subject scores) ship in `squad-friends.ts` and `study-data.ts`, and a Dataverse table `cree0_student` is wired for Power Platform persistence.

TopicData

id / subjectId identity
memoryScore number | null (not started)
lastReviewedAt ISO date
nextReviewAt ISO date, 9:00 AM
quizAttempts count

QuizRoomDraft

roomId / hostUserId identity
subject / topic what's being rescued
roundNumber of totalRounds
status 'active' | 'finished'
questionStartedAt timer anchor

RescueNudgeLog

memberId / senderName who nudged whom
subject / topic the weak node
rescueStatus pending | completed | expired
roomId linked rescue room

QuizRoomParticipantDraft

displayName player identity
avatarColor Yellow | LightBlue | White
score live points
lastAnswerCorrect round feedback

SquadMember

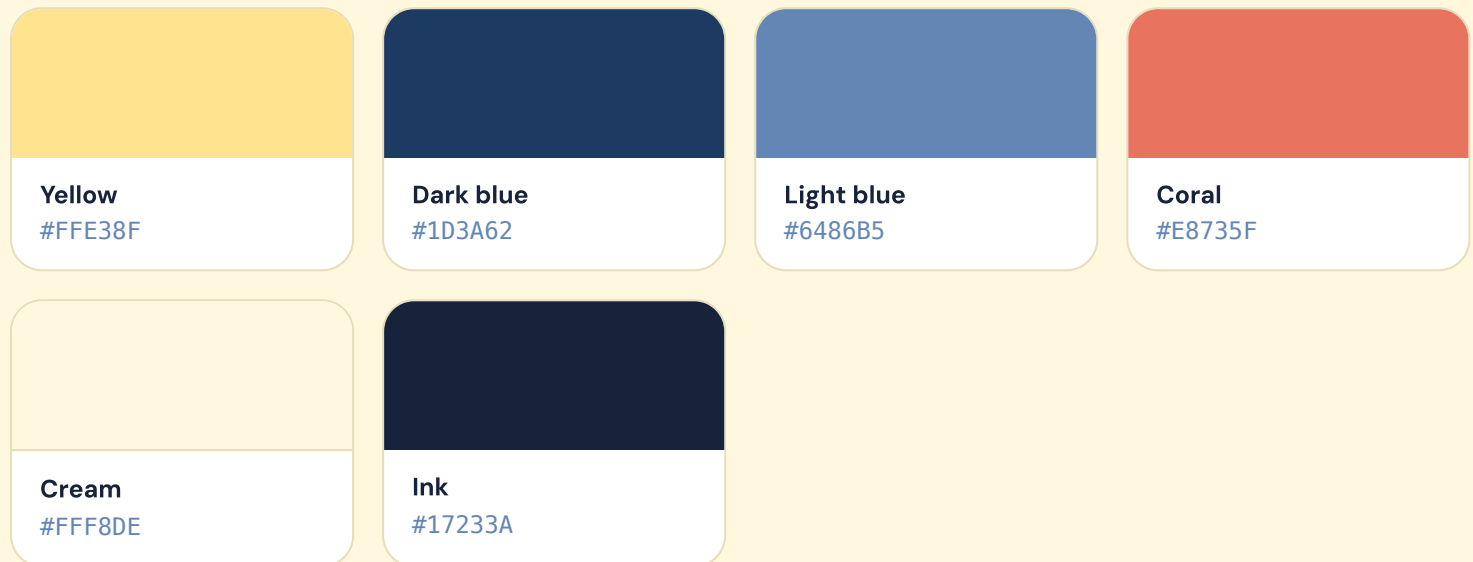
name e.g. Maya Tan
overallMemoryScore squad ranking
subject scores per-topic breakdown

Question

format mcq | structured | essay
difficulty foundation → advanced
markScheme points + keywords
keywordHints wrong → correct + why

Warm, floaty, and unmistakably navy.

The "modern warm startup" theme is defined once in `src/index.css` as CSS custom properties, then mapped into Tailwind v4 tokens. This document is set in the same system: Anton for display, DM Sans for body, 1.25rem radii, floaty cards with soft navy shadows, and hard offset shadows for hero moments.



Type & shape

- **Anton** — display headings and the EDUNETS. wordmark
- **DM Sans** — body, UI and data (Canva Sans first in-app)
- Radius scale from `--radius: 1.25rem` ; cards at 18px
- Pill buttons, chip badges in sidebar blue `#DDE9F8` , coral uppercase eyebrows

Signature moves

- `.floaty-card` — lift 3px on hover with a deepening navy shadow
- `.edunets-gradient` — radial yellow/blue washes over cream
- Hard offset shadow (`18px 18px 0 #1D3A62`) on the hero tile
- Web motif — concentric rings and spokes echo Spidey's concept web everywhere from onboarding progress to empty states

Two targets, one bundle.

Web (Vercel)

- Push the repo to GitHub → import in Vercel → auto-detected Vite build (`npm run build` → `dist/`)
- Preview at `/welcome`, then promote it to the root route in a separate release, keeping the dashboard at `/dashboard`
- Secrets stay out of the repo — environment variables live in Vercel project settings

Power Platform

- `power.config.json` registers the app ("StudyNow app") with its Dataverse instance and `cree0_student` data source
- `initialize()` from `@microsoft/power-apps` boots the bridge on app start
- Same `dist/` bundle serves both targets

Known limits of the launch slice

The current build runs on demo data: onboarding answers persist only in the browser session, and the material-upload and voice-note flows shown in onboarding still need a storage/transcription backend before accepting real student files. Multiplayer rescue state is drafted client-side pending a realtime service.